

Ahnaf Atef Choudhury

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience in building scalable applications, microservices, and SQL/NoSQL databases, with expertise in secure cloud deployments on AWS and Azure using Docker and Kubernetes.

TECHNICAL SKILLS

- **Programming and Frameworks:** Java, Spring Boot, Python, Angular, TypeScript, Node.js
- **APIs and Integrations:** REST, GraphQL, XML, microservices architecture
- **Databases:** PostgreSQL, MySQL, MongoDB, Oracle, Cassandra, Snowflake, Redis
- **Cloud and DevOps:** AWS, Azure, GCP, Docker, Kubernetes, Helm
- **CI/CD, Testing and Others:** Jenkins, GitHub Actions, Junit, Terraform (IaC), Git, Maven, Gradle, Agile/Scrum

EXPERIENCE

Software Engineer | Optum, Washington, DC July 2024 - Present

- Refactored Optum's QuickConnect, a secure file exchange portal built with Angular 18 and Spring Boot microservices, enhancing client usability and reducing average page load time by 40 percent.
- Designed and maintained Java Spring Boot microservices and versioned REST and GraphQL APIs handling over 2 million daily requests with 99.99 percent uptime across provider reimbursement and claims platforms.
- Optimized PostgreSQL, MySQL, and MongoDB schemas through indexing and query tuning which reduced reporting latency by 30 percent for high volume transactions.
- Implemented event driven workflows using Apache Kafka and RabbitMQ to manage real time claim updates and ensure reliable asynchronous communication between distributed microservices.
- Leveraged AWS EKS, EC2, S3, RDS, Lambda, DynamoDB, SNS, SQS, and CloudWatch for deployment, scaling, and monitoring which improved reliability through alerting and automated performance dashboards.
- Established observability practices by integrating structured logging, metrics, and tracing with Splunk and CloudWatch to strengthen incident response and root cause analysis.
- Automated unit and regression testing with JUnit and integrated coverage reports into GitHub Actions and Jenkins CI/CD pipelines which reduced production defects and improved release stability.
- Deployed containerized Java services using Docker and Kubernetes across AWS and Azure using Helm charts for orchestration and blue green deployments to maintain zero downtime releases.

NLP Engineer | 22nd Century Technologies, Mclean, VA May 2023 – March 2024

- Designed backend services in FastAPI and deployed them in Docker and Kubernetes clusters to improve scalability and uptime.
- Built document processing pipelines combining Snowflake for large-scale reporting and MongoDB for unstructured metadata, managing 10M+ records powering real-time compliance dashboards.
- Integrated LLMs and NLP models using Hugging Face, Llama, OpenAI API, LangChain, and LlamaIndex to classify 100K+ policy documents with 92% accuracy in production.
- Implemented retrieval-augmented generation (RAG) workflows and used vector databases such as Pinecone to improve context-aware document search and classification.
- Collaborated with data scientists, AI engineers, and automation specialists to fine-tune models and integrate UiPath bots and Orchestrator APIs with NLP pipelines for automated document handling.

Java Fullstack Developer | MetLife Bangladesh, Dhaka, Bangladesh January 2019 – June 2021

- Developed Java Spring Boot and Angular applications that supported 300,000 monthly policyholders with 99.9% uptime across core insurance platforms.
- Enhanced PostgreSQL and MongoDB databases by applying partitioning strategies and query optimization which cut reporting latency by 35% for claims processing systems.
- Automated CI/CD pipelines using Jenkins reducing deployment cycles from 2 hours to 25 minutes per release.
- Deployed containerized Java services on AWS using Docker and Kubernetes ensuring reliable and scalable production environments.
- Implemented web services using REST, XML, and SQL to integrate policy and claims data across downstream systems, ensuring reliable interoperability and compliance with insurance standards.

EDUCATION

MS in Applied IT | George Mason University, USA August 2021 – May 2024

BS in Computer Science and Engineering | Brac University, Bangladesh January 2015 – December 2018